

Yufeng (Alex) Xia

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RESEARCH INTERESTS

- MSc by Research student at the Institute for Computing Systems Architecture (ICSA), University of Edinburgh, supervised by Prof. Mahesh K. Marina. My research interests lie at the intersection of **AI systems, networked systems, and large-scale infrastructure emulation**. My work spans **AI-RAN**, where live radio processing and data-driven workloads share the same computing infrastructure under real-time constraints, as well as **emulation and digital twin systems** that provide scalable platforms for developing and validating RAN and edge-AI algorithms. My research has led to a paper conditionally accepted by **ACM MobiCom 2026**.
- **Availability:** MSc by Research thesis submits September 2026. Available remotely from **September 2026**, full-time (remote or in Hong Kong) from **December 2026**. Applying for **PhD entry in Fall 2027** and to the **Hong Kong PhD Fellowship Scheme**.

EDUCATION

- **University of Edinburgh — MSc by Research, Computer Science** Edinburgh, UK
Supervised by Prof. Mahesh K. Marina. Thesis: Flux-DT: An Elastic, Preemption-Tolerant Digital Twin Runtime for AI-RAN Sep 2025 – Sep 2026 (expected)
- **University of Edinburgh — BSc (Hons) Computer Science** Edinburgh, UK
*Grade: **First Class Honours** (weighted average 74/100; UK First Class $\geq 70/100$)* Sep 2021 – May 2025

SELECTED PUBLICATIONS

- EmuRAN: Scalable and Cost-Effective Cloud-based RAN Emulation System** *MobiCom'26 (cond. accepted)*
*Ujjwal Pawar, Andrew E. Ferguson, **Yufeng Xia**, Xenofon Foukas, Mahesh Marina, Bozidar Radunovic*
- Morphling: Emulator for Distributed Machine Learning at the Edge** *EdgeSys'26 (Best Paper)*
*Leyang Xue, **Yufeng Xia**, Eren Mendi, Ismaeel Bashir, Jiaxun Yang, Myungjin Lee, Mahesh Marina*
- Weaver: Foundation Model Training over AI-RAN Compute Infrastructure** *under submission*
*Leyang Xue, Tianxin Wang, Xin Zhe Khooi, Jiaxun Yang, Dheeraj Mahendiran, **Yufeng Xia**, Mun Choon Chan, Myungjin Lee, Mahesh Marina*
- Wasp: Foundation Model Training System for the Edge** *SEC'26*
*Leyang Xue, **Yufeng Xia**, Myungjin Lee, Mahesh Marina*
- CoSenseAQ: Compiler Optimizations using Sensor Technical Specifications with Automatic Quantization** *under submission*
***Yufeng Xia**, Pei Mu, Antonio Barbalace*

RESEARCH AND PROJECT EXPERIENCE

- **Weaver — Foundation Model Training over Shared AI Compute Infrastructure** Under Submission
Research Assistant; GPU profiling, interference characterisation, controller evaluation 2024 – 2026
 - **Harvesting accelerators from a latency-critical tenant:** Evaluated the controller that decides how much accelerator capacity an AI workload may harvest from infrastructure shared with a latency-critical tenant, a near-real-time resource-allocation problem on GPU-accelerated hardware. On a multi-site testbed it harvests up to **83%** of the available spare compute without degrading the co-tenant's performance.
 - **Roofline analysis and deadline measurement:** Ran the GPU profiling and roofline analysis showing the priority workload's kernels are memory-bandwidth-bound while training is compute-bound, the asymmetry that makes SM-level co-location possible; instrumented its scheduler to measure per-deadline completion, and carried out the large-scale distributed training runs used in the evaluation.
- **Morphling — Emulator for Distributed Machine Learning at the Edge** EdgeSys 2026
*Research Assistant; **Best Paper Award**; owned the C++ transport data plane* Oct 2025 – Mar 2026

- **Inter-device transport:** Built the event-driven (libevent) proxy client/server carrying tensor traffic between emulated devices: **2,400 lines of C++** in an 18k-line C++/CUDA codebase. Emulation fidelity is bounded by this path, so transfer cost had to be driven to the microsecond level.
- **Serialisation cost and distributed correctness:** Measured serialisation overhead across Protobuf and FlatBuffers along the full send path and chose the batching strategy from the measurements; implemented virtual-clock synchronisation across emulated devices and fixed the connection-cleanup and device-failure deadlocks that appeared only at scale.
- **Flux-DT — Elastic, Preemption-Tolerant Runtime for Stateful Workloads on Reclaimable Compute** MSc by Research Thesis
University of Edinburgh — sole author Sep 2025 – Present
 - **State that must survive preemption:** Existing compute-harvesting designs assume the harvesting workload is stateless and disposable. Built a runtime for a workload that is neither: it holds state that must survive interruption, so it can run on bursty, reclaimable capacity without violating the priority tenant's guarantees or corrupting its own state.
 - **Near-real-time control policy:** Designed and implemented a control loop that answers resource pressure with a graded set of actions (relocation, fidelity reduction, suspend and resume) rather than a single eviction decision, on the timescale the priority workload imposes.
 - **Scheduling heuristic and shedding order:** Formulated work placement as a joint objective across interfering units and **designed a greedy heuristic** to decide which work is worth doing; the same ranking determines the shedding order when compute is reclaimed mid-solve.
- **EmuRAN — Scalable, Cost-Effective Cloud-based RAN Emulation System** MobiCom 2026
Research Assistant; owned the end-to-end evaluation Conditionally Accepted
 - **Cloud deployment at scale:** Deployed and operated the emulator on public cloud across **130 instances and 6,400 CPU cores**, emulating 10,000 mobile devices and 10,000 base stations, two orders of magnitude larger than existing software solutions.
 - **Fidelity and scalability evaluation:** Designed and ran the evaluation campaign: fidelity against real hardware (random-access timelines, iperf3 throughput, per-slot completion-time distributions) and scalability under node count and constrained compute, establishing the system behaves correctly rather than merely running at scale.
- **CoSenseAQ — LLVM Auto-Quantization from Sensor Specifications** Under Submission
Research Assistant, ICSA; first author Apr 2024 – May 2025
 - **Compiler-level optimisation:** Built an LLVM auto-quantization framework driven by information a compiler cannot recover from source alone, using variable-range analysis and type shrinkage to reduce code size and execution time on embedded targets; low-level C/C++ and cross-compilation toolchain work throughout.
 - **Measurement on real hardware:** Evaluated on ARM Cortex-M0 and Cortex-A53, measuring power, performance and area. Drove the work from problem formulation through implementation and evaluation to manuscript.

TECHNICAL SKILLS

- **Languages** C++ , C, Python, Java, Bash.
- **AI/ML & Systems** PyTorch, distributed foundation-model training ,resource management, CUDA GPU programming and profiling, workload scheduling, large-scale system emulation and benchmarking .
- **Distributed system and Cloud deployment** Multi-site deployment, Kubernetes, Docker, distributed virtual-time synchronisation
- **Networking and Wireless** 5G RAN experimentation with OpenAirInterface and Open5GS, RAN emulation and network digital twins, throughput/latency measurement .
- **Engineering** Git and code review, CMake, technical documentation (see attached toolchain and repository summaries).

INDUSTRY EXPERIENCE

- **Senior Tech — Back-end Development Intern** Suzhou, China
Production backend APIs in a commercial engineering team; live platform migration from JFinal to Spring Boot, code review and release testing. Java, MySQL, Redis, Kafka. Jun 2023 – Sep 2023

AWARDS, TEST SCORES AND SERVICE

- **Awards Best Paper**, EdgeSys 2026 (9th Workshop on Edge Systems, Analytics and Networking, co-located with ACM MobiSys 2026), for *Morphling*. **Student Travel Grant**, ACM MobiSys 2026.
- **Test Scores GRE 336/340** (Verbal 166, Quantitative 170/170, Jun 2024); IELTS 7.0.
- **Service** Student volunteer, EuroSys 2026 (on-site conference operations).
- **Languages** Mandarin (native), English (fluent).

REFERENCES

- **Prof. Mahesh K. Marina** — MSc by Research supervisor, School of Informatics, University of Edinburgh · mahesh@ed.ac.uk
- **Dr Antonio Barbalace** — BSc dissertation supervisor and Junior Research Assistant supervisor, School of Informatics, University of Edinburgh · antonio.barbalace@ed.ac.uk